Introduction to R Programming

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Introduction to R

- **R** is a language designed for data analysis, statistics, and modeling.
- A tool throughout the first year: econometrics, problem sets, and eventually research.
- **Advantages:** flexible, powerful, easy to start using, and fully open-source.
- **What you can do quickly:**
  - load and explore data,
  - compute summary statistics,
  - make plots,
  - solve linear algebra problems.
- **Our goal:** build confidence with the basics.
  - Focus on understanding core ideas rather than memorizing syntax.
  - Learn how R stores data, how to pass objects into functions, and how to interpret the output.

Working in RStudio

RStudio is the main interface used in practice.
It is organized into four main panels, each with a different purpose:

- **Console** — where code runs. Commands typed here execute immediately.
- **Script Editor** — where you write and save your `.R` scripts. You send code from here to the Console.
- **Environment / History** — shows the objects currently in memory (vectors, data frames, functions) and your command history.
- **Plots / Files / Packages / Help** — displays plots, your working directory, installed packages, and documentation.

If you do not have R or RStudio installed, you can download them from:

- R: https://cran.r-project.org
- RStudio: https://posit.co/download/rstudio-desktop/

Today's Roadmap

- Basic objects: vectors, matrices, data frames.
- tidyverse essentials: pipes, `summarise()`, `group_by()`.
- Simple plotting with `ggplot2`.
- Control flow: `if`, loops, and apply functions.
- Writing functions in R.
- RMarkdown: Markdown is a formatting syntax for authoring HTML, PDF, and MS Word documents.